A FINITE ALGEBRA OVER A HENSELIAN PAIR WITH FINITE CLOSED FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Henselian pair and the lifting of idempotents. the statement counted by the sibling card S1-algebra-lift-idempotent at 1784-1803. Supplies the splitting of the finite algebra.
- Finiteness of an algebra over a ring and its preservation under passing to a factor. used at proof lines 2302-2330. Keeps the pieces finite over the pair.
- Finiteness of the closed fibre and its behaviour under the decomposition. used at proof lines 2330-2344. Supplies the condition that the pieces have to inherit.

Source reference. Original lemma 0EM0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18290–18290. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07M4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9714–9714. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00H5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 631–631. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05CK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6923–6923. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00I1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 663–663. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let (A, I) be a Zariski pair. Let $A \to B$ be a finite ring map. Assume*

1. *$B/IB = B_1 \times B_2$ is a product of A/I-algebras*
2. *$A/I \to B_1/IB_1$ is surjective,*
3. *$b \in B$ maps to $(1, 0)$ in the product.*

Then there exists a monic $f \in A[x]$ with $f(b) = 0$ and $f \bmod I = (x-1)x^d$ for some $d \geq 1$.

Proof. By Lemma 07M4 we can find an étale ring map $A \to A'$ inducing an isomorphism $A/I \to A'/IA'$ such that $B' = B \otimes_A A'$ contains an idempotent e' lifting the image of b in B'/IB'. Consider the corresponding A'-algebra decomposition

$$B' = B'_1 \times B'_2$$

which is compatible with the one given in the lemma upon reduction modulo I. The map $A' \to B'_1$ is surjective modulo IA'. By Nakayama's lemma (Algebra, Lemma 00DV) we can find $i \in IA'$ such that after replacing A' by A'_{1+i} the map $A' \to B'_1$ is surjective. Observe that the image $b'_1 \in B'_1$ of b satisfies $b'_1 - 1 \in IB'_1$. Thus we may pick $a' \in IA'$ mapping to $b'_1 - 1$. On the other hand, the image $b'_2 \in B'_2$ of b is in IB'_2. By Algebra, Lemma 00H5 there exist a monic polynomial $g(x) = x^d + \sum a'_j x^j$ of degree d with $a'_j \in IA'$ such that $g(b'_2) = 0$ in B'_2. Thus the image $b' = (b'_1, b'_2) \in B'$ of b is a root of the polynomial $(x - 1 - a')g(x)$. We conclude that

$$(b' - 1)(b')^d \in \sum\nolimits_{j=0,\ldots,d} IA' \cdot (b')^j$$

We claim that this implies

$$(b-1)b^d \in \sum\nolimits_{j=0,\ldots,d} I \cdot b^j$$

in B. For this it is enough to see that the ring map $A \to A'$ is faithfully flat, because the condition is that the image of $(b-1)b^d$ is zero in $B/\sum_{j=0,\ldots,d} Ib^j$ (use Algebra, Lemma 05CK). The map $A \to A'$ flat because it is étale (Algebra, Lemma 00U2). On the other hand, the induced map on spectra is open (see Algebra, Proposition 00I1 and use previous lemma referenced) and the image contains $V(I)$. Since I is contained in the Jacobson radical of A we conclude. □